

has benefitted majorly from technology. The IoT has made it easy to monitor health. It helps transmit signals directly to hospitals, making it easy to track in real time and provide necessary medication immediately. Thanks to the IoT, portable and handheld health monitoring devices, and moving healthcare units will be soon available across all parts of India.

Autonomous vehicles

From parking sensors to driverless cars, the IoT has helped evolve the automobile industry. Vehicles now come with an inbuilt GPS system that helps drivers navigate a new place or drive in a different country. Devices are now installed in vehicles that can track driver behaviour and, in case of an accident, notify the family.

GPS lets vehicles be tracked via both computer and smartphones. This lets people know where their vehicles are at all times.

Developments in the field of electric cars are happening fast, to find the best solutions for EV batteries, their charging stations, battery recycling and so on. We may expect electric cars running on the road by 2020.

Toll taxes can now be paid by smart sensing systems or smart cards. This removes the need for standing in long queues. Such electronic toll collection systems are based on RFID or GPS.

Flying cars, hyperloops, high-speed trains and much more are under development.

Mobile technology

It is believed that by 2020, every individual will own 10 gadgets. This itself has added to the economy.

Payments and transactions can be made with the swipe of a finger. This makes it easier to track goods and take necessary action, if required.

Benefits of 5G such as high Internet speed, lower latency and higher capacities will soon be common. This will make 5G a prime technology trend in 2020 and beyond.

Technologies for fast-charging batteries and wireless charging for smartphones would become a regular and affordable feature soon. Wire-



AR for construction industry (Credit: <https://readwrite.com>)

less charging offers opportunities for automotive, healthcare and other industries.

Foldable, flexible and rollable displays for wearables devices, tablets and laptops are also in development stages. QLED, OLED and IGZO are the latest technologies that will soon find applications in all TVs, mobile screens, tablets and other gadgets.

Robots, cobots and drones

These have been widely adopted in the military, medical and manufacturing industries. Cobots have eliminated the risks of working with robots. Many personalised robots have also been developed to assist humans in most areas. These are assisting human beings for making decisions based on facts.

Robots will soon be seen empowering law and order at banks and police stations. Countries are working towards developing a number of machines that can benefit police departments.

Latest technology in robotics is integrating robots with AI so that they can interact with humans in a better way.

Robotic fish SoFi is helping explore the underwater world.

From healthcare, e-commerce and retail to agriculture and defence, drone technology is being utilised for surveillance and services (delivery). Drones are being widely used for military applications, too. But soon, their commercial use will be seen in areas

like photography, recording, construction site monitoring and the like.

Drones provide precise and accurate information about the work in progress. These also save a lot of time, as entire sites can be surveyed in just a few minutes.

As drone technology is rapidly developing accuracy and precision in its readings, even less human involvement will be required. Governments across the world are also working towards keeping registration of drones and robots official.

Nanoscience and nanotechnology

Nanotechnology is the application of nanoscience that deals with the use of nanomaterials and nano-sized components in useful products. It provides the ability to design custom-made materials and products with new and enhanced properties. It has a wide range of application in nano-electronic components, medicines and sensors, and interfaces between electronic and biological systems, automotive, nanoelectronics devices and more.

Scientists and engineers are deliberately making nanomaterials to take advantage of their enhanced properties such as lighter weight, higher strength, increased control of light spectrum and greater chemical reactivity. Dealing at nanoscale enables them to utilise the unique physical, mechanical, chemical and optical properties of such materials.